



PC4x4SAR | Tech Preparedness Quick Reference

piercecounty4x4sar.org | Download full guide at piercecounty4x4sar.org/safety

EMERGENCY

CALL 911

NAVIGATION APPS

- ❑ **AllTrails+**: \$36/yr. Best for trail discovery + offline maps. iOS & Android.
- ❑ **Gaia GPS**: \$40/yr. Best for serious backcountry. Topo maps. Used by SAR teams.
- ❑ **onX Backcountry**: \$30/yr. Best for land ownership + hunting. Offline topo.
- ❑ **CalTopo**: \$50/yr. Best for trip planning. Custom topo maps + route analysis.
- ❑ **Organic Maps**: FREE. Open-source offline maps. Hiking trails + elevation. No account.

FREE OFFLINE MAPS

- ❑ **Google Maps**: Search area > 3-dot menu > Download offline map. Expires in 30 days.
- ❑ **Apple Maps**: Profile > Offline Maps > Download New Map. iOS 17+ required.
- ❑ **Organic Maps**: Open app, zoom to area, auto-downloads. 100% free, no account.

BATTERY CONSERVATION

- ❑ **Low Power Mode**: Enable at trailhead. iPhone: Settings > Battery. Android: Settings > Battery.
- ❑ **Brightness**: Reduce to 30-40% -- screen is the #1 battery drain.
- ❑ **Background apps**: Close social media and unused apps before hiking.
- ❑ **Camera batches**: Take photos in bursts, then lock phone immediately.
- ❑ **Airplane mode**: ONLY if you have a separate satellite communicator. Otherwise leave cellular ON.

POWER BANKS

- ❑ **Day hike**: 5,000 mAh (1 recharge, 4-5 oz)
- ❑ **Overnight**: 10,000 mAh (2-3 recharges, 6-8 oz)
- ❑ **Multi-day**: 20,000 mAh (4-5 recharges, 12-14 oz)
- ❑ **Cold weather**: Keep bank in inside pocket near body. Cold cuts capacity 20-40%.

SATELLITE EMERGENCY

- ❑ **FREE 911 Texting**: T-Mobile/Starlink -- all carriers, 60+ phones. Sign up before trip at t-mobile.com/satellite-911
- ❑ **iPhone 14+ SOS**: Built-in satellite SOS. Practice at: Settings > Emergency SOS
- ❑ **Android SOS**: Pixel 9+, Galaxy S25+ have satellite emergency messaging.
- ❑ **Garmin inReach**: Two-way satellite texts + SOS. ~\$15/mo. Best for regular use.
- ❑ **Zoleo**: Two-way satellite + cell + WiFi messaging. ~\$20/mo.

PRE-TRIP CHECKLIST

- ❑ Phone at 100%
- ❑ Power bank at 100%
- ❑ Offline maps downloaded + tested
- ❑ Medical ID updated
- ❑ Emergency contacts set
- ❑ Satellite SOS signed up

CELL COVERAGE -- CHECK BEFORE YOUR TRIP

Most Cascade mountain trails and Mt. Rainier backcountry have NO cell service. Check your carrier's coverage map before heading out.



T-Mobile

<https://www.t-mobile.com/coverage>



AT&T

<https://www.att.com/maps/wireless-coverage>



Verizon

<https://www.verizon.com/coverage-map>